

Vatsal Omar

Software Engineer — AI & Messaging Systems

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Software engineer specializing in AI-driven messaging systems and real-time notification workflows. Built production-grade omni-channel communication platforms using Firebase Cloud Messaging, WebSockets, and multi-agent orchestration. Experienced in designing personalized, low-latency messaging solutions for enterprise customers.

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani

2023-08 – 2026-07

Bachelor of Science in Computer Science — Pilani, IN

EXPERIENCE

Software Development Intern

2024-12 – 2025-09

BLive — Bangalore, IN

- Architected sub-second notification routing system using Firebase Cloud Messaging and Python backend to deliver personalized SOS alerts, delivery updates, and anomaly notifications to operator roles
- Developed real-time messaging infrastructure with WebSockets and Go for AcordLayer, enabling low-latency threaded communication and workflow automation for privacy-conscious teams
- Optimized Flutter app performance by 25% using DevTools profiling, supporting high-frequency telemetry updates in mission-critical notification workflows

PROJECTS

AcordLayer — Self-Hosted Collaboration Platform

Go · TypeScript · WebSockets · PostgreSQL · Docker

Real-time messaging platform with workflow automation and AI extensibility for privacy-focused teams

- Built WebSocket-based messaging system with threaded channels and real-time notifications in Go, handling concurrent multi-user collaboration
- Implemented containerized workflow automation layer with Docker Compose for on-premises and cloud deployments

ChaosOps AI — Multi-Agent Incident Response Simulator

Python · FastAPI · TRL/GRPO · OpenAI · Anthropic

Hierarchical multi-agent system for simulating production failure resolution with rogue-agent detection

- Created LLM-agnostic agent framework with function-calling and streaming parsing across OpenAI, Anthropic, and open-source providers
- Developed deterministic incident replay system to reproduce failure scenarios with (failure_type, seed) traceability

ACHIEVEMENTS

Top 8 — OpenEnv AI Hackathon — Ranked top 8 out of 70,000+ developers in India's largest AI hackathon (Meta × Hugging Face × PyTorch), building RL environments on Meta's OpenEnv

Intern of the Quarter — BLive — Recognized for technical excellence in notification workflow development and cross-team collaboration

TECHNICAL SKILLS

Languages: TypeScript · Python · Go · Dart · SQL

Frontend: Flutter · React · Next.js 14 · Tailwind CSS

Backend: FastAPI · NestJS · PostgreSQL · Firebase · Redis

Ai MI: Multi-agent orchestration · Prompt Engineering · Function-calling · GRPO/RL · LLM integration

